

SWYNEX BUSINESS INTELLIGENCE CASE STUDY

Hospital Management Analytics

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Internship Domain: Data Analytics

Project Title: Hospital Management Analytics Dashboard

Tools Used: Microsoft Power BI

Project Type: Business Intelligence Case Study

Dashboard Tool: Microsoft Power BI

Key Analysis: Patient demographics, hospital performance, department-wise patient distribution, admission types, and billing analysis.

1. Project Overview

The Hospital Management Analytics Dashboard is a Business Intelligence project developed using Microsoft Power BI. This project analyzes hospital-related data and presents meaningful insights through interactive visualizations, charts, KPIs, and filters.

The dashboard helps users understand patient demographics, hospital performance, department-wise patient distribution, admission types, and billing information. It supports data-driven decision-making and improves the understanding of hospital operations.

2. Business Problem

Hospitals generate large amounts of data related to patients, admissions, departments, and billing. Analyzing this information manually can be time-consuming and make it difficult to identify important patterns.

A centralized dashboard is needed to summarize hospital data, compare performance, and identify useful business insights.

This project addresses the problem by transforming hospital data into an interactive Power BI dashboard.

3. Business Objectives

The main objectives of this project are:

- Analyze the total number of patients.
- Understand the average age of patients.
- Monitor the total hospital billing amount.
- Compare patient distribution across departments.
- Analyze patients by gender and blood type.
- Compare admission types and hospital-wise patient counts.
- Provide interactive filters for exploring hospital data.
- Support data-driven operational decisions.

4. Tools and Technologies Used

Microsoft Power BI: Used to create the dashboard and data visualizations.

Power Query: Used for data preparation and transformation, where required.

Data Modeling: Used to organize tables and establish relationships between datasets.

DAX (Data Analysis Expressions): Used to create measures and calculate key performance indicators, where applicable.

Hospital Dataset: Used as the source of patient, admission, department, and billing information.

5. Dashboard Design and Key Performance Indicators

The dashboard is titled Hospital Management Analytics and contains KPI cards, charts, and interactive slicers.

The main KPIs displayed are:

1. Total Patients: 10K
2. Average Patient Age: 51.57
3. Total Billing Amount: 270.58M

The dashboard includes the following visualizations:

- Patients by Gender: Doughnut chart showing gender distribution.

- Patients by Blood Type: Pie chart showing blood type distribution.
- Patients by Department: Bar chart comparing department-wise patient counts.
- Patients by Admission Type: Column chart comparing admission categories.
- Patients by Hospital: Column chart comparing patient counts across hospitals.
- Hospital Filter: Allows users to select individual hospitals.
- Department Filter: Allows users to explore specific departments.
- Patient Name Filter: Helps users explore selected patient records.

These visualizations provide a consolidated overview of hospital data in one report.

6. Key Business Insights

Insight 1: Patient Volume

The dashboard displays approximately 10,000 patients, providing an overview of the patient population represented in the dataset.

Insight 2: Average Patient Age

The average patient age is 51.57 years. This helps users understand the age profile of the recorded patient population.

Insight 3: Billing Analysis

The total billing amount displayed is approximately 270.58 million in the dataset's currency units. This KPI helps summarize the total billing value represented by the available records.

Insight 4: Gender and Blood Type Distribution

The gender and blood type charts show the distribution of patients across different categories. These visualizations help users understand the composition of the patient population.

Insight 5: Department and Hospital Comparison

The department-wise and hospital-wise charts allow users to compare patient counts across departments and hospitals. These comparisons can help identify areas that may require further investigation or resource planning.

7. Interactive Features

The dashboard provides interactive features that allow users to explore hospital data efficiently.

- Hospital slicer to filter records by hospital.
- Department slicer to filter records by medical department.
- Patient name filter to explore individual patient records.
- Interactive charts that filter related visualizations when selected.
- KPI cards that summarize important hospital metrics.

These features make the dashboard user-friendly and help users focus on specific information.

8. Business Recommendations

Based on the dashboard's analytical capabilities, the following recommendations can be considered:

1. Resource Planning: Review department-wise patient counts to support staffing and resource allocation.
2. Billing Monitoring: Regularly review billing indicators and investigate unusual changes.
3. Patient Demographics: Analyze age and gender distributions to support service planning.
4. Hospital Performance: Compare patient volumes across hospitals and investigate significant differences.
5. Admission Analysis: Examine admission categories to understand hospital service demand.
6. Data Quality: Regularly check for missing values, duplicates, and inconsistent records to improve reporting accuracy.

These recommendations should be validated using additional operational information before implementation.

9. Project Limitations

The quality of the insights depends on the accuracy and completeness of the available dataset. The dashboard summarizes recorded information but does not independently establish the reasons for differences between hospitals or departments.

Additional data about staffing, treatment outcomes, operating costs, and historical trends could provide deeper insights in future versions.

10. Conclusion

The Hospital Management Analytics Dashboard demonstrates how Microsoft Power BI can transform hospital data into an interactive Business Intelligence report.

By presenting key performance indicators, patient demographics, department-wise distribution, hospital comparisons, and billing information, the dashboard makes hospital data easier to understand.

This project helped develop practical skills in data visualization, dashboard design, data modeling, and business analysis. It demonstrates the value of Business Intelligence in supporting informed decision-making.

11. Project Deliverables

- Power BI dashboard file (.pbix).
- Business Intelligence case study document.
- GitHub repository containing the project files.
- Dashboard screenshot for project presentation.

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